AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT101
General Mathematics

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 17/Nov. /2025

Number of Exam Pages: 2 (including this cover sheet)	**Time Allowed: 90 minutes**

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Instructions:

- Total Marks:**60 marks**
- This exam consists of **4 questions**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

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ESSAY QUESTIONS

Each question is worth 15 marks. You should answer all the questions, giving 60 possible marks.

Q–1: [3+6+6 Marks]

a) write the answer as a + bi (complex number in standard form)

i) find f(1+i) if f(x)= (x+1)(x-1).

ii) $\frac{3i}{4-5i}$.

b) Solve for x: 3(5x -1) +5(3x+2) = 7

Q–2: [8+7Marks] solve the following:

a) | 2x- 2 | < 4

b) $7x - 1 > x + 11$ *or* $-11x > -33$ (write the solution in interval notation).

Q–3: [6+7+2] Marks] a) solve the following equations:

i) $\sqrt[3]{x + 40}$= -5

ii) x^2-6x - 3=0 (use the quadratic formula)

b) Find the domain of f(x)= $\sqrt{x - 3}$

Q–4: [8+7Marks] Given the two points A (0,6) and B (4, 9).

a) Find the distance between the two points A and B and the midpoint of the line segment AB.

b) Find the equation of the line that passes through the two points A and B.

End of the questions.

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